

Aeration Basin Retrofit at Greenfield Water Reclamation Plant Design Report, Revision C

The Greenfield Water Reclamation Plant treats wastewater for a service area of 48,000 residents. Its two aeration basins were commissioned in 1987 and have operated continuously since. Dissolved oxygen in Basin 2 has fallen below the permitted floor of 2 milligrams per litre on 31 days in the past year, each time during the summer peak. The plant operator recorded the longest single excursion as 9 hours.

The cause is not the blowers, which were replaced in 2019 and still meet their rated output. It is the diffuser grid on the basin floor. The original coarse-bubble diffusers transfer oxygen at roughly 1.2 kilograms per kilowatt-hour under clean water conditions, and the membranes have fouled unevenly, so a third of the grid now delivers most of the air. Short-circuiting carries that air to the surface before it dissolves.

Left alone, the plant will breach its discharge permit during the next heat wave. The regional authority has issued a notice of concern, and a second excursion in the same calendar year would trigger a formal compliance review. This report sets out the retrofit options and recommends one.

Constraints

Four constraints shaped every option below, and they are listed in the order that ruled candidates out.

The plant cannot be taken offline. There is no third basin and no equalisation storage, so one basin must remain in service at all times. Any retrofit has to be installed one basin at a time, with the working basin carrying the full load for the duration. The plant's hydraulic model shows a single basin can carry peak flow for no more than 14 consecutive days before solids carry over into the clarifiers.

The budget is 1.4 million dollars, approved in the current capital cycle. Funds not committed by the end of the fiscal year return to the general reserve and would have to be requested again.

The existing basin floor cannot carry additional dead load. A 1994 structural assessment limits any new fixed equipment on the floor slab to 40 kilograms per square metre.

Finally, the retrofit must not increase the plant's electrical demand. The site's service connection is already at 92 percent of its rated capacity, and upgrading it would require a new transformer and a separate utility agreement that takes 18 months.

Options considered

Replacing the diffuser grid with fine-bubble membrane diffusers is the recommended option. Fine-bubble units transfer oxygen at about 2.8 kilograms per kilowatt-hour, more than double the present grid, which both restores the dissolved oxygen margin and lowers blower demand. Installation fits the 14-day single-basin window, and the units mount on the existing floor supports at 22 kilograms per square metre.

Mechanical surface aerators were ruled out on electrical demand. Four units sized for Basin 2 would add 150 kilowatts, pushing the site past its service capacity and triggering the transformer upgrade and its 18-month utility agreement.

A third aeration basin was ruled out on both cost and schedule. The lowest construction estimate received was 3.2 million dollars, more than twice the approved budget, and the earliest completion date was 26 months away. It remains the right answer if the service area grows past 60,000 residents, and the site plan reserves land for it.

Chemical oxygen dosing was ruled out as an ongoing cost rather than a fix. It would meet the permit floor during excursions but adds an estimated 180,000 dollars a year in reagent, against a one-off retrofit, and does nothing about the fouled grid.